

B2M
(30420) | v1.5

Failure Mode and Effects Analysis (FMEA)

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Product:	NEXUS B2M (30420)
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1. Purpose
This FMEA evaluates potential failure modes in the manufacturing process of NEXUS B2M (Reference 30420) Beta-2 Microglobulin immunoassay kits. The analysis highlights assembly and changeover risks following shared production line usage with other NEXUS products.

2. Scope
Applicable to the complete NEXUS B2M manufacturing process at Riverside. Shared assembly line AL-003 is used for multiple NEXUS products including NEXUS CK MB and NEXUS Troponin.

3. FMEA Risk Assessment

Process Step Controls		D	Failure Mode RPN	Effect	S	Cause	O
Kit_Assembly Clearance checklist, operator training	3	54	Incomplete line clearance	Wrong component in kit	9	Rushed changeover	2
Kit_Assembly 3 Visual verification, barcode scan	4	108	Wrong sensor cartridge type	Cross-contamination between products	9	Short changeover time	
CART_Receipt 2 Temperature monitoring, incoming QC	3	42	CART coating degradation	Reduced assay sensitivity	7	Storage temperature excursion	
Strip_Assembly Daily dispenser verification	3	42	Incorrect reagent volume	Assay inaccuracy	7	Dispenser calibration drift	2

Calibration Calibrator reconstitution error Invalid calibration curve 8 Operator technique variation 2

Process Step	D	Failure Mode RPN	Effect	S	Cause	O	Controls
training, single-use QC_Testing		Contaminated control sample 3 30	False QC failure	5	Improper sample handling	2	Aseptic technique aliquots
Labeling label verification system	2 18	Wrong product label applied	Product mix-up at customer site	9	Label roll changeover error	1	Camera-based
Packaging verification, checklist	3	Missing insert / IFU 30	Customer unable to perform test	5	Packaging station error	2	Weight

4. Risk Priority Summary
Critical threshold: RPN >= 80. One failure mode exceeds the critical threshold:
- Wrong sensor cartridge type (RPN=108): This is the highest-priority risk. Assembly line changeover procedure SOP-ASM-003 mandates a minimum 30-minute changeover time with an 8-item clearance checklist.
Barcode scanning of sensor cartridge sachets is required before kit assembly begins.

Moderate risks (RPN 50-80):
- Incomplete line clearance (RPN=54): Mitigated by operator training and clearance checklist per SOP-ASM-003.
- Calibrator reconstitution error (RPN=64): Visual aids and standardised training programme in place.

5. Revision History

Version	Date	Author	Changes
1.0	2023-01-10	L. Reynolds	Initial release
1.5	2024-06-01	L. Reynolds	Added wrong sensor cartridge type risk (RPN=108); updated changeover controls

